

IoT Fundamentals - Chapter 1

The Internet of Things (IoT) refers to the network of physical objects embedded with sensors, software, and connectivity to exchange data. IoT systems typically consist of sensors, actuators, connectivity modules, and cloud platforms. Key protocols include MQTT, CoAP, and HTTP. Applications range from smart homes and wearable devices to industrial automation and smart cities.

Key components of an IoT system include: 1) Sensors and actuators for data collection, 2) Microcontrollers like Arduino and Raspberry Pi, 3) Communication protocols such as Wi-Fi, Bluetooth, Zigbee, and LoRa, 4) Cloud platforms for data storage and analytics.

IoT Fundamentals - Chapter 2

MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol designed for constrained devices and low-bandwidth networks. It uses a publish-subscribe model where clients connect to a broker. Topics are hierarchical strings used to filter messages. QoS levels: 0 (at most once), 1 (at least once), 2 (exactly once).

CoAP (Constrained Application Protocol) is designed for resource-constrained devices. It uses UDP instead of TCP, making it lighter than HTTP. CoAP supports GET, PUT, POST, DELETE methods similar to REST APIs. It features built-in discovery and multicast support.

IoT Fundamentals - Chapter 3

Security in IoT is critical due to the massive number of connected devices. Common threats include: device tampering, eavesdropping, man-in-the-middle attacks, and DDoS attacks. Best practices include: encryption (TLS/SSL), secure boot, firmware updates, and access control.

Edge computing processes data closer to where it is generated rather than sending everything to the cloud. Benefits include reduced latency, bandwidth savings, and improved privacy. Popular edge platforms include AWS Greengrass, Azure IoT Edge, and Google Cloud IoT.